KS3 DIGITAL TECHNOLOGY

ETHICAL HACKING AND CAPTURE THE FLAG

Think like an attacker to defend like a pro: the law and ethics of hacking, how Capture the Flag competitions work, decoding hidden messages, finding secrets in files and web pages, and how ethical hackers report problems and build careers.

LESSONS

1 Hackers with permission
2 Inside a Capture the Flag
3 Codes and encodings
4 Hidden in plain sight
5 Web detective
6 Report it, don't exploit it

LESSON 1 OF 6

HACKERS WITH PERMISSION

Ethical hackers test systems only with clear permission and within agreed limits, then report what they find so it can be fixed.

STARTER · 5 MINUTES

DO NOW

A friend shows you how to log in to another pupil's account "just to prove it's possible". Is that hacking? Is it a crime?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain the difference between ethical and criminal hacking and the rules of the Computer Misuse Act

SUCCESS CRITERIA: I CAN...

- ✓ I can describe white, black and grey hat hackers
- ✓ I can explain the three main offences in the Computer Misuse Act
- ✓ I can explain why permission and scope matter

THREE HATS

- White hat: ethical hackers who test with permission to make systems safer
- Black hat: criminals who break in to steal, damage or show off
- Grey hat: break in without permission, then tell the owner, which is still illegal
- Only one of these is a career

KEY WORDS

ETHICAL HACKING	Testing computer systems for weaknesses with the owner's permission, so they can be fixed.
PENETRATION TESTING	A planned, authorised attack on a system to find its weaknesses.
COMPUTER MISUSE ACT	The UK law (1990) that makes unauthorised access to computers and spreading malware criminal offences.
SCOPE	The agreed limits of a security test: which systems and methods are allowed.

THE COMPUTER MISUSE ACT 1990

- Applies across the UK, including Northern Ireland
- Offence 1: unauthorised access to computer material (e.g. using someone's password)
- Offence 2: unauthorised access with intent to commit further crimes
- Offence 3: unauthorised acts to impair a computer (e.g. spreading malware)
- Penalties range from fines to life imprisonment for the most serious cases

PERMISSION AND SCOPE

- Ethical hackers get written permission before testing
- The scope says exactly which systems and methods are allowed
- Stay inside the scope; report everything; damage nothing
- In this topic we only ever hack practice systems made for it

ACTIVITY · 15 MINUTES · PAIRS

LEGAL OR NOT?

1. Read the eight scenario cards (e.g. "A pupil guesses a teacher's password to change a mark").

2. Decide for each: legal ethical hacking, or a Computer Misuse Act offence? If it's an offence, which one?

3. For each illegal scenario, write what the person should have done instead.

In your workbook, answer:

? Choose two illegal scenarios: which offence, and what should the person have done?

? Explain why a grey hat hacker can still be breaking the law.

Your workbook: Lesson 1 · tick each step off as you go

QUICK CHECK

1. What makes hacking "ethical"?

A Nothing is changed
B The owner has given permission and it's done within agreed limits
C It's done by a teenager
D It's done at weekends

2. Which law makes unauthorised access to computers illegal in the UK?

A The Copyright Act
B The Online Gaming Act
C The Highway Code
D The Computer Misuse Act 1990

3. Logging in to a friend's account without permission "just to look" is...

A only against school rules
B an offence under the Computer Misuse Act
C legal for under-16s
D legal if you change nothing

4. What is the scope of a penetration test?

A A telescope
B The length of a password
C The number of hackers
D The agreed list of systems and methods allowed

QUICK CHECK: ANSWERS

1. **B: The owner has given permission and it's done within agreed limits**
Permission is the key difference.

2. **D: The Computer Misuse Act 1990**
The CMA covers hacking.

3. **B: an offence under the Computer Misuse Act**
Unauthorised access is a crime.

4. **D: The agreed list of systems and methods allowed**
Scope sets the limits.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe the three main offences in the Computer Misuse Act.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe white, black and grey hat hackers
- ✓ I can explain the three main offences in the Computer Misuse Act
- ✓ I can explain why permission and scope matter

LESSON 2 OF 6

INSIDE A CAPTURE THE FLAG

Capture the Flag competitions are legal hacking games.

STARTER · 5 MINUTES

DO NOW

You find a note that says "The answer is hidden in plain sight." What would you do first?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how CTF competitions work and use a structured approach to solve challenges

SUCCESS CRITERIA: I CAN...

- ✓ I can describe jeopardy and attack-defence CTFs
- ✓ I can name common CTF challenge categories
- ✓ I can use a structured method to solve a challenge

WHAT IS A CTF?

- A legal competition where you hack systems built for the game
- Find a "flag": a secret string like FLAG{h1dd3n_1n_pla1n_s1ght}
- Submit flags to score points
- Played by pupils, students and professionals worldwide

KEY WORDS

CAPTURE THE FLAG	A legal hacking competition where players find hidden "flags" in challenges built for the purpose.
FLAG	A secret text string hidden in a CTF challenge, which you submit to score points.
CYBERCHEF	A free browser tool made by GCHQ for encoding, decoding and analysing data.

TYPES AND CATEGORIES

- Jeopardy CTF: a board of challenges, more points for harder ones
- Attack-defence CTF: defend your own server while attacking others' (advanced)
- Categories: cryptography, forensics, web, OSINT, reverse engineering, puzzles
- Teams split up challenges by their strengths

A METHOD FOR EVERY CHALLENGE

- Read the title and description carefully: they're full of hints
- Look at everything: files, text, source code, file names
- Recognise patterns: what format is this?
- Try, note what you tried, ask for a hint if stuck

ACTIVITY · 30 MINUTES · TEAMS OF 3

FIRST FLAGS

1. Log in to the practice CTF your teacher has set up.
2. Choose roles: a reader (reads challenges and hints), a solver (tries ideas) and a note-taker.
3. Solve as many starter challenges as you can in 25 minutes. Rotate roles after each flag.
4. Record each flag and the method you used.

In your workbook, answer:

? Write down one flag you found and explain step by step how you found it.

? What did your team do when you got stuck?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What is a flag in a CTF?

A A secret string you find and submit to score points
B A piece of cloth
C A computer virus
D A team name

2. In a jeopardy CTF...

A everyone answers the same question
B teams choose challenges from a board of categories
C teams fight with water guns
D only one person plays

3. What does OSINT stand for?

A Open-source intelligence
B Online secret internet
C Optical sensor input
D Operating system integration

4. Who made CyberChef?

A GCHQ
B The BBC
C NASA
D Apple

QUICK CHECK: ANSWERS

1. **A: A secret string you find and submit to score points**
   Flags prove you solved a challenge.

2. **B: teams choose challenges from a board of categories**
   It's a board of puzzles.

3. **A: Open-source intelligence**
   Publicly available information.

4. **A: GCHQ**
   It's GCHQ's free tool.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe what a Capture the Flag competition is.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe jeopardy and attack-defence CTFs
- ✓ I can name common CTF challenge categories
- ✓ I can use a structured method to solve a challenge

LESSON 3 OF 6

CODES AND ENCODINGS

Encodings change data into a different format so it can be stored or sent, but anyone can reverse them.

STARTER · 5 MINUTES

DO NOW

What do these have in common? 01001000 01101001 / 48 69 / SGk= / Uv. (Hint: they all say the same thing.)

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Recognise and decode binary, hexadecimal, Base64 and ROT13, and explain the difference between encoding and encryption

SUCCESS CRITERIA: I CAN...

- ✓ I can recognise binary, hex, Base64 and ROT13
- ✓ I can decode them using CyberChef
- ✓ I can explain the difference between encoding and encryption

ENCODING IS NOT ENCRYPTION

- Encoding: changing data's format so it can be stored or sent (no secret key)
- Anyone who knows the encoding can reverse it
- Encryption: scrambling data with a secret key so only key-holders can read it
- CTFs often hide flags in layers of encoding

KEY WORDS

ENCODING	Changing data into a different format for storage or sending, with no secret key.
BASE64	An encoding that turns data into letters, digits, + and /, often ending in =.
HEXADECIMAL	A number system with 16 digits (0-9 and A-F), often used to show bytes.
ROT13	A simple cipher that shifts each letter 13 places; doing it twice gives the original.

SPOT THE FORMAT

- Binary: only 0s and 1s, often in groups of 8
- Hex: digits 0-9 and letters A-F, often in pairs
- Base64: upper and lower case letters, digits, + and /, often ending in = or ==
- ROT13: looks like normal words, but scrambled; shift letters by 13

USE CYBERCHEF

- Paste the text into the Input box
- Drag in a recipe: "From Base64", "From Hex", "ROT13"
- Try "Magic": it guesses the encoding for you
- Layered? Stack several steps in one recipe

ACTIVITY · 25 MINUTES · TEAMS OF 3

DECODING RELAY

1. Your teacher gives each team a sheet of ten encoded messages.
2. For each, identify the encoding from how it looks before decoding it.
3. Decode it using CyberChef (or by hand for ROT13 and binary).
4. The last three are layered: you'll need more than one step.

In your workbook, answer:

? For two messages, write the encoding you spotted and the clue that told you.

? Explain why Base64 shouldn't be used to protect a password.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is the main difference between encoding and encryption?

A Encryption uses a secret key; encoding doesn't
B They are the same
C Encryption only works on numbers
D Encoding is more secure

2. Which text is most likely Base64?

A 01100001
B Uryyb
C U2VjcmV0IGZsYWc=
D 4A 6F

3. Which characters appear in hexadecimal?

A Only symbols
B Only 0 and 1
C All letters A-Z
D 0-9 and A-F

4. What does applying ROT13 twice do?

A Doubles the shift to 26 and scrambles it
B Returns the original text
C Turns it into binary
D Deletes it

QUICK CHECK: ANSWERS

1. **A: Encryption uses a secret key; encoding doesn't**
   Encoding can be reversed by anyone.

2. **C: U2VjcmV0IGZsYWc=**
   Mixed case, digits and = padding.

3. **D: 0-9 and A-F**
   Hex is base 16.

4. **B: Returns the original text**
   13 + 13 = 26, a full circle.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why encoding a password in Base64 does not keep it secure.

Look back at today's success criteria: can you tick them all?

- ✓ I can recognise binary, hex, Base64 and ROT13
- ✓ I can decode them using CyberChef
- ✓ I can explain the difference between encoding and encryption

LESSON 4 OF 6

HIDDEN IN PLAIN SIGHT

Steganography hides messages inside ordinary-looking files, such as in the pixels of an image.

STARTER · 5 MINUTES

DO NOW

How could you hide a secret message in a photo so nobody even knows a message exists?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how data can be hidden in files and images, and investigate files for hidden information

SUCCESS CRITERIA: I CAN...

- ✓ I can explain steganography and how it differs from encryption
- ✓ I can find information in file metadata
- ✓ I can use file signatures to identify a file's real type

STEGANOGRAPHY

- Hiding a message so nobody knows it's there
- Ancient version: invisible ink, messages under wax on writing tablets
- Digital version: tiny changes to pixel colours that the eye can't see
- Encryption hides what a message says; steganography hides that it exists

KEY WORDS

STEGANOGRAPHY	Hiding a message inside something ordinary so nobody knows it exists.
METADATA	Hidden information stored with a file, such as its date, author or location.
FILE SIGNATURE	The first bytes of a file, which identify its real file type.
DIGITAL FORENSICS	Recovering and investigating evidence from computers and other devices.

DATA ABOUT DATA

- Metadata: hidden information stored with a file
- Photos: date, camera, settings and sometimes GPS location
- Documents: author name, edit history
- Investigators (and CTF players) always check metadata

WHAT IS THIS FILE REALLY?

- The file extension (.jpg, .pdf) is just part of the name: it can lie
- A file signature ("magic number") is in the first bytes of the file
- PNG files start with the bytes 89 50 4E 47 (".PNG")
- Renaming a file doesn't change what it really is

ACTIVITY · 25 MINUTES · PAIRS

FORENSICS CHALLENGE

1. Download the three challenge files from the practice CTF.
2. File 1: check its metadata for a hidden comment.
3. File 2: check its file signature and fix its extension.
4. File 3: use the steganography tool your teacher shows to reveal a hidden message in an image.

In your workbook, answer:

? Which flag was hardest to find, and what technique did you use?

? Explain how a file can pretend to be a different type, and how you can tell.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is steganography?

A Scrambling a message with a key
B Hiding a message so nobody knows it exists
C Compressing a file
D Writing in shorthand

2. How can a message be hidden in an image?

A By making the image bigger
B By changing the file name
C By making tiny changes to pixel colours
D By printing it

3. What is a file signature?

A The file's size
B The file name
C The author's handwritten signature
D Bytes at the start of a file that show its real type

4. Which information might photo metadata reveal?

A The colour of the camera
B The viewer's password
C Where and when it was taken
D Nothing at all

QUICK CHECK: ANSWERS

1. **B: Hiding a message so nobody knows it exists**
   It conceals existence, not just meaning.

2. **C: By making tiny changes to pixel colours**
   Changes in the lowest bits are invisible.

3. **D: Bytes at the start of a file that show its real type**
   Magic numbers identify file types.

4. **C: Where and when it was taken**
   Metadata can include GPS and time.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between encryption and steganography.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain steganography and how it differs from encryption
- ✓ I can find information in file metadata
- ✓ I can use file signatures to identify a file's real type

LESSON 5 OF 6

WEB DETECTIVE

Everything a browser receives can be seen: HTML source, comments, scripts and cookies.

STARTER · 5 MINUTES

DO NOW

Right-click a web page and choose "View page source". What's the strangest thing you can find?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Investigate a practice website using its source code and explain how developers protect websites

SUCCESS CRITERIA: I CAN...

- ✓ I can view a page's source and find comments and hidden elements
- ✓ I can explain what cookies and robots.txt are
- ✓ I can explain why secrets must never be sent to the browser

YOUR BROWSER SEES EVERYTHING

- View source: the HTML the server sent
- Developer tools (F12): inspect elements, scripts, network requests and cookies
- Comments <!-- like this --> don't show on the page but are in the source
- Hidden elements are only hidden from view, not from you

KEY WORDS

SOURCE CODE	The code that makes a program or web page, such as HTML, CSS and JavaScript.
COOKIE	A small piece of data a website stores in your browser, for example to keep you logged in.
DEVELOPER TOOLS	Tools built into web browsers for inspecting a page's code, network requests and storage.
SERVER	A computer that provides services or web pages to other computers.

COOKIES AND ROBOTS.TXT

- Cookies: small pieces of data a website stores in your browser
- They remember logins, settings and shopping baskets
- robots.txt: a public file telling search engines what not to list
- Ironically, it can point attackers to "secret" pages

HOW DEVELOPERS DEFEND

- Never put secrets (passwords, keys) in HTML or JavaScript
- Check permissions on the server, every time
- Validate all input; never trust anything from the browser
- Regular testing, including paid ethical hackers

ACTIVITY · 30 MINUTES · PAIRS

WEB DETECTIVE CHALLENGES

1. Open the practice website your teacher gives you (a CTF site built for this).
2. Find flags hidden in: an HTML comment, a hidden element, a JavaScript file, a cookie and the robots.txt file.
3. For each flag, write one rule a developer should have followed to avoid the problem.

In your workbook, answer:

? List the flags you found and where each was hidden.

? For two of them, write the developer rule that would have prevented it.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. How can you see a web page's HTML?

A View page source or developer tools
B You can't
C Turning off the monitor
D Printing the page

2. Why are HTML comments a bad place for secrets?

A They're encrypted
B Anyone can read them in the source code
C They're only visible to admins
D They're deleted automatically

3. What is a cookie?

A A snack for hackers
B A virus
C A type of password
D A small piece of data a website stores in your browser

4. What should developers do with input from the browser?

A Trust it completely
B Delete it
C Validate it on the server, never trusting it
D Print it

QUICK CHECK: ANSWERS

1. **A: View page source or developer tools**
   Browsers let you see source code.

2. **B: Anyone can read them in the source code**
   Comments are public.

3. **D: A small piece of data a website stores in your browser**
   Cookies remember settings and logins.

4. **C: Validate it on the server, never trusting it**
   Attackers can change anything client-side.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why hiding a page by not linking to it is not real security.

Look back at today's success criteria: can you tick them all?

- ✓ I can view a page's source and find comments and hidden elements
- ✓ I can explain what cookies and robots.txt are
- ✓ I can explain why secrets must never be sent to the browser

LESSON 6 OF 6

REPORT IT, DON'T EXPLOIT IT

When ethical hackers find a weakness, they report it privately so it can be fixed (responsible disclosure).

STARTER · 5 MINUTES

DO NOW

You notice your school's website shows other pupils' details if you change a number in the web address. What should you do?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe responsible disclosure and bug bounties, and explore careers in cyber security

SUCCESS CRITERIA: I CAN...

- ✓ I can describe responsible disclosure
- ✓ I can explain what a bug bounty is
- ✓ I can describe routes into cyber security careers

FOUND A PROBLEM? REPORT IT

- A vulnerability is a weakness that could be exploited
- Responsible disclosure: tell the owner privately and give them time to fix it
- Don't test further, don't share it, don't use it
- At school: tell a teacher or the IT team straight away

KEY WORDS

VULNERABILITY	A weakness in a system that could be exploited to cause harm.
RESPONSIBLE DISCLOSURE	Privately reporting a vulnerability to its owner and giving them time to fix it before telling anyone else.
BUG BOUNTY	A reward paid to people who responsibly report vulnerabilities.

BUG BOUNTIES

- Companies pay rewards to people who report vulnerabilities properly
- Run by big tech firms, banks and governments
- Strict rules: only test what's in scope, never harm users or data
- Some ethical hackers earn a living from bug bounties

CAREERS IN CYBER

- Penetration tester, security analyst, incident responder, digital forensics investigator, security engineer
- Belfast and Northern Ireland are home to many cyber security companies
- Routes: GCSE and A level, apprenticeships, degrees, CyberFirst schemes
- Start now: CTFs, CyberFirst competitions and coding clubs

ACTIVITY · 25 MINUTES · PAIRS

WRITE A DISCLOSURE REPORT

1. Read the scenario: while using a (fictional) cinema booking website, you notice changing the booking number in the address shows someone else's booking.

2. Write a short responsible disclosure email to the cinema: what you found, how they can see it, why it matters, and what you did and didn't do.

3. Swap with another pair and check: is it clear, polite and responsible?

In your workbook, answer:

? Write your disclosure email.

? Explain why you should not test the problem further or post about it online.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is responsible disclosure?

A Privately reporting a vulnerability so it can be fixed
B Selling a weakness
C Posting a weakness online
D Ignoring a weakness

2. What is a bug bounty?

A A fine for hackers
B A prize for finding insects
C A type of virus
D A reward for responsibly reporting a vulnerability

3. You accidentally see other pupils' data on a school system. What should you do?

A Post about it
B Look at more records to check
C Stop and tell a teacher or IT straight away
D Show your friends

4. What does a penetration tester do?

A Sells computers
B Writes viruses
C Legally attacks systems to find weaknesses
D Repairs pens

QUICK CHECK: ANSWERS

1. **A: Privately reporting a vulnerability so it can be fixed**
   Report privately, give time to fix.

2. **D: A reward for responsibly reporting a vulnerability**
   Companies pay for reports.

3. **C: Stop and tell a teacher or IT straight away**
   Stop and report.

4. **C: Legally attacks systems to find weaknesses**
   Pen testers are ethical hackers.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe the steps an ethical hacker should follow after finding a vulnerability.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe responsible disclosure
- ✓ I can explain what a bug bounty is
- ✓ I can describe routes into cyber security careers

END OF TOPIC

ETHICAL HACKING AND CAPTURE THE FLAG

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.